

REFERME INVESTMENTS

Functional Requirements Document (FRD)

Investment Analytics, Scoring & Power BI Decision Layer

Document Control	Details
Document ID	RMI-FRD-001
Project	Investment Analytics & Decision-Support Framework
Project Period	February - June 2026
Version	1.0
Status	Portfolio
Methodology control	Scoring weights, thresholds and technical implementation details require stakeholder approval before production use.

1. Purpose

Translate the approved business requirements into functional behavior for investment-target data management, KPI calculation, scoring, recommendation support, Power BI analysis, drill-down, portfolio monitoring, risk alerts and data-quality visibility.

2. Functional Scope

- Investment-target master and evaluation context
- KPI calculation and normalization
- Five-dimension scoring and overall Investment Score
- Recommendation support
- Five Power BI pages
- Business filtering/drill-down
- Portfolio trend monitoring
- Risk and data-quality visibility

3. Functional Requirements

ID	Functional Requirement
FR-01	Maintain investment-target master data including business ID/type and evaluation attributes.
FR-02	Calculate KPI values from validated financial, operational, customer and risk inputs.
FR-03	Normalize KPI values to a common scoring scale according to approved thresholds.
FR-04	Calculate five dimension scores using approved KPI weights.
FR-05	Calculate overall Investment Score using approved dimension weights.
FR-06	Display investment pipeline KPIs including businesses evaluated, average score and decision counts.
FR-07	Provide individual Investment Evaluation view with overall and dimension scores.
FR-08	Provide Business Profile trends for revenue, customers, profitability, operations, growth and risk.
FR-09	Provide Portfolio Monitoring for actual vs target, trends, score changes and early warnings.
FR-10	Provide Risk & Alerts view for revenue decline, churn, margin deterioration, concentration and cash-flow performance.
FR-11	Allow filtering/drill-down by investment target and relevant business dimensions.
FR-12	Expose data-quality status to prevent false confidence in incomplete evaluations.

4. Detailed Functional Behaviour

Function	Inputs	Processing / Rule	Output
Target setup	Business ID, business type, evaluation attributes	Validate required identifiers and maintain target context	Investment-target master
KPI calculation	Validated financial, operational, customer and risk inputs	Apply approved KPI definitions/formulas	Applicable KPI values
KPI normalization	KPI value + approved threshold/benchmark	Convert to comparable scoring scale	Normalized KPI score
Dimension scoring	Normalized KPI scores + approved KPI weights	Calculate weighted result per dimension	Five dimension scores
Investment scoring	Five dimension scores + approved dimension weights	Weighted 0-100 calculation	Investment Score
Recommendation	Investment Score + approved decision thresholds + risk rules	Apply approved decision logic	Invest / Review / Reject
Data-quality control	Missing, invalid, duplicate or stale input indicators	Calculate/derive data-quality status	Data Quality Score / confidence status

Monitoring	Actuals, targets, historical KPI/score/risk values	Compare periods and targets	Trend, variance and attention list
Alerts	Approved threshold + current KPI/risk condition	Detect breach and identify affected target	Risk / early-warning alert

5. User Stories & Acceptance Criteria

ID	User Story	Acceptance Criteria
US-01	As an investment analyst, I want to view all businesses with their investment scores so that I can prioritize due diligence.	Given evaluated businesses exist, when I open the overview, then I can see businesses evaluated, average score and decision counts.
US-02	As an underwriter, I want to drill into a target's five dimension scores so that I can understand the drivers of the recommendation.	When a business is selected, then the overall score, five dimension scores and underlying KPIs are visible.
US-03	As a portfolio manager, I want actual-vs-target and score trends so that I can detect post-investment deterioration.	For an invested business, the monitoring view shows actual vs target, KPI trend, score change and risk change.
US-04	As a risk reviewer, I want alerts for declining revenue, high churn, margin deterioration and concentration so that I can investigate early.	When an approved threshold is breached, the affected business and indicator are highlighted for review.
US-05	As a decision maker, I want data-quality visibility so that I can distinguish weak performance from weak evidence.	When required inputs are incomplete or invalid, the Data Quality Score/status is visible alongside the evaluation.

6. Power BI Functional Requirements

Page	Decision Question	Required Content
1. Investment Overview	What does the investment pipeline look like?	Businesses Evaluated; Average Investment Score; Invest/Review/Reject; Average Risk, Growth and Financial Strength
2. Investment Evaluation	Should ReferMe invest in this business?	Overall score; five dimension scores; underlying KPIs
3. Business Profile	What is happening inside this target?	Revenue trend, customers, profitability, operations, growth and risk
4. Portfolio Monitoring	How are investments performing after investment?	Actual vs target; KPI trends; score/risk changes; early warnings; attention list
5. Risk & Alerts	Where should the investment team investigate?	Revenue decline, churn, margin deterioration, concentration and strong cash-flow signals

7. Data Requirements

Data Domain	Representative Fields	Used For
Business Master	Business ID, type, evaluation status	Target identification and filtering
Financial	Revenue, gross profit, EBITDA, operating cash flow, costs	Financial Strength / Risk
Workforce & Operations	Employees, labor cost, capacity, on-time service	Operational Performance
Customers	Customer count, new/lost/repeat customers, complaints, customer revenue	Growth / Customer Health / concentration
Time	Month/quarter/year	Trends, CAGR, volatility, actual vs target
Scoring	KPI score, dimension score, investment score, recommendation	Decision support
Data Quality	Missing, invalid, duplicate or stale input indicators	Data Quality Score / confidence

8. KPI Catalogue

Dimension	KPI	Business Meaning / Calculation Intent
Financial Strength	Revenue	Scale of business / top-line performance
Financial Strength	Revenue Growth %	Change in revenue over comparison period
Financial Strength	Gross Margin %	Gross profit as % of revenue
Financial Strength	EBITDA Margin %	EBITDA as % of revenue
Financial Strength	Operating Cash Flow	Cash generated from operations
Financial Strength	Cash Flow Margin %	Operating cash flow as % of revenue
Operational Performance	Revenue per Employee	Revenue / employee count
Operational Performance	Operating Cost %	Operating costs as % of revenue
Operational Performance	Labor Cost %	Labor cost as % of revenue
Operational Performance	Capacity Utilization %	Used capacity / available capacity
Operational Performance	On-Time Service %	On-time services / total services
Operational Performance	Operational Efficiency Score	Composite operating-efficiency measure
Growth	Customer Growth %	Change in active customer base
Growth	New Customer Acquisition	New customers acquired in period
Growth	Recurring Revenue %	Recurring revenue / total revenue
Growth	Average Revenue per Customer	Revenue / active customers
Growth	12-Month Revenue CAGR	Annualized revenue growth
Customer Health	Customer Retention %	Retained customers / eligible prior customers
Customer Health	Customer Churn %	Lost customers / eligible prior customers
Customer Health	Repeat Customer Rate	Repeat customers / total customers
Customer Health	Customer Complaint Rate	Complaints / customer or service base
Customer Health	Customer Concentration %	Revenue exposure to largest customer(s)
Risk	Revenue Volatility	Variability of revenue over time
Risk	Customer Concentration Risk	Risk derived from customer dependency
Risk	Margin Deterioration	Negative movement in margin
Risk	Cash Flow Risk	Risk arising from weak/unstable operating cash flow
Risk	Data Quality Score	Completeness/validity/consistency of required data
Risk	Overall Business Risk Score	Composite business-risk measure

9. Scoring Configuration

Dimension	Proposed Weight
Financial Strength	25%
Operational Performance	20%
Growth	20%
Customer Health	15%
Risk	20%
Total	100%

10. Error / Exception Handling

Condition	Expected Functional Behaviour
Mandatory input missing	Flag data-quality limitation; do not present unsupported confidence.
Invalid denominator / non-calculable KPI	Return KPI as not calculable and surface the reason.
Scoring weights do not total 100%	Block/flag score configuration until corrected.
Decision thresholds not approved	Display score and supporting evidence without presenting the recommendation as production-approved.
Critical risk condition	Surface risk prominently and support manual review / override logic once approved.
No historical period	Do not fabricate trend/CAGR/volatility; indicate insufficient history.

11. UAT Mapping

Test ID	Scenario	Test Data / Action	Expected Result
UT-01	KPI calculation	Load complete valid test data	All applicable KPIs calculate with no unexpected blanks
UT-02	Dimension scoring	Provide KPI scores and approved weights	Dimension score equals weighted KPI result
UT-03	Investment scoring	Provide five dimension scores	Overall score equals approved weighted calculation
UT-04	Recommendation	Use test scores around approved thresholds	Correct Invest/Review/Reject label returned
UT-05	Drill-down	Select one investment target	Underlying dimension/KPI values reconcile to summary
UT-06	Portfolio trend	Load multiple periods	Actual-vs-target and score/risk changes display correctly
UT-07	Risk alert	Create threshold-breach scenario	Relevant warning appears for affected business
UT-08	Business profile	Select target	Financial, customer, operations, growth and risk trends filter correctly
UT-09	Data quality	Remove/invalid required inputs	Data-quality status declines and limitation is visible

12. Traceability Summary

Business Requirement	Functional Requirement	User Story / Test	Output
BR-01	FR-04	US-02 / UT-02	Investment Evaluation
BR-02	FR-02	US-02 / UT-01	Investment Evaluation / Business Profile
BR-03	FR-03, FR-04	US-02 / UT-02	Dimension scoring
BR-04	FR-05	US-01, US-02 / UT-03	Overview / Evaluation
BR-05	FR-05	US-01 / UT-04	Recommendation
BR-06	FR-07, FR-11	US-02 / UT-05	Drill-down
BR-07	FR-09	US-03 / UT-06	Portfolio Monitoring
BR-08	FR-10	US-04 / UT-07	Risk & Alerts
BR-09	FR-08, FR-11	US-02 / UT-08	Business Profile
BR-10	FR-12	US-05 / UT-09	Data-quality visibility